KUKU AI GENERALIST ASSIGNMENT - SOLUTION DOC

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Problem 1: Find the Promos | Two-Stage Hybrid Extraction Pipeline

Repo: github.com/sanketjambhule/kuku-promo-tool

Prototype: Streamlit Community Cloud

1. PROBLEM FRAMING & STRATEGIC CONTEXT

Microdrama platforms (Kuku, ReelShort, DramaBox) scale through continuous, high-volume ad creative testing. User acquisition hinges entirely on 20 to 60-second vertical promotional video ads posted across TikTok, Reels, and Shorts. Today, human editors watch full episodes to manually cut promos - a bottleneck that breaks when releasing multiple shows weekly.

We framed promo extraction as an Attention Economics Optimization problem across three non-negotiables:

- **The 0-3s Hook:** Must arrest passive infinite scrolling with immediate conflict, a physical slap, or an outrageous accusation.
- **The Final 3s Cliffhanger:** Must cut precisely at peak unresolved emotional tension, forcing viewers to click to Episode 1.
- **Standalone Comprehension:** Cold scrolling audiences with zero prior lore must instantly understand who the victim and villain are.

2. TWO-STAGE HYBRID PIPELINE ARCHITECTURE

Brute-force LLM evaluation of all sliding windows in a 2-minute episode (~250 windows) costs ~$1.50/ep and takes 60s+. Across a 100-episode season, this is financially unsustainable. We designed a Two-Stage Hybrid Engine:

```
[Screenplay .txt] -> [Parser] -> [Sliding Windows 20-60s] -> [Stage 1 Heuristic + NMS] -> Top 10 -> [Stage 2 LLM] -> [Ranker]
```

Stage 1: Deterministic Heuristic Filter (<50ms execution in Python)

- Beat Extraction & Timestamp Normalization: Parses scenes, character dialogue, actions, and camera cues.
- Missing Timestamp Interpolation: Automatically estimates speech rate (2.5 words/sec, min 2s) if timestamps are missing.
- Virality Lexicon Scoring: Scans S-Tier Drama (slap, marry, betray, kill, bastard), A-Tier (liar, kneel, revenge), B-Tier (desperate).
- Hook & Cliffhanger Structural Check: Rewards dialogue/action opening in first 4s and unresolved endings (?, !, ...).
- Temporal Non-Maximum Suppression (NMS): Suppresses windows with IoU overlap > 65% to guarantee diverse scene coverage.

Stage 2: Semantic LLM Evaluator (Single Batched Call via Pydantic Schema)

The filtered Top 10 candidate windows are passed in a single structured prompt to Google Gemini 1.5 Flash or Claude 3 Haiku. The model analyzes subtle character status dynamics, emotional escalation, and outputs strict Pydantic JSON.

3. MULTI-CRITERIA COMPOSITE RANKING FORMULA

Viral Score = 2.0 x [(0.40 x Hook Rating) + (0.40 x Cliffhanger Rating) + (0.20 x Standalone Comprehension Score)]

Ratings range 1-5. Standalone Comprehension maps: High=5.0, Medium=3.0, Low=1.0. Multiplied by 2.0 to scale onto 1.0 - 10.0.

4. KEY ASSUMPTIONS MADE

- **Screenplay-Video Fidelity:** Assumes final video edit reflects screenplay dialogue sequence and timestamps.
- **Pacing Baseline:** For un-timed screenplays, dialogue rate is calibrated to 2.5 words/second with a 2.0s floor for non-verbal actions.
- **Duration Boundary (20-60s):** Clips under 20s lack dramatic setup; clips over 60s suffer severe retention drop-off on TikTok/Shorts.

5. POSTING STRATEGY: WHICH PROMOS TO POST FIRST & WHY?

Running the pipeline across all 50 episodes yielded 214 ranked promos. We established a 3-tier marketing deployment strategy:

POST #1: Primary Paid User Acquisition Ad (Meta & TikTok Ads) -> Episode 1 [00:21.80 -> 01:21.20] (Score: 9.20/10)
Hook: Matriarch stops bride: 'You should turn around, Juliana' | Cliffhanger: Juliana begs father: 'Father... tell them this is a mistake.'
Why First: High Standalone Comprehension. Cold paid traffic will not tolerate complex backstories; a bride humiliated at the altar for her cruel sister creates instant emotional outrage that directly drives app installs.

POST #2: Organic Viral Hook (TikTok / Reels Algorithmic Loop) -> Episode 42 [00:18.00 -> 00:39.90] (Score: 8.80/10)
Hook: Rupert shrieks: 'Guards! Arrest this imposter!' | Cliffhanger: Royal Guard: 'We will not... because we know who he is.'
Why Second: 5/5 Hook Shock Value. Sudden defiance of authority triggers massive organic replay loops and comment debates, driving free viral reach even among casual non-subscribers.

POST #3: Mid-Funnel Retargeting (Cliffhanger Conversion) -> Episode 34 [00:13.10 -> 00:44.80] (Score: 8.60/10)
Hook: Witchcraft accusation in dock | Cliffhanger: Maid Prudence steps onto witness stand weeping to give coerced false testimony.
Why Third: Maximum cliffhanger urgency (5/5). Converts warm users who watched an earlier ad teaser without completing the install.

6. QUALITY CHECKING: WHERE DOES THE TOOL PICK BADLY?

- **Subtle Irony & Deadpan Sarcasm:** Keywords flag overt conflict. A whisper-delivered line like 'How delightful to see you, sister' scores lower in Stage 1 keyword density despite being a brilliant psychological hook.
- **Visual-Only Action Beats:** A silent slap or poison-drop described in CAMERA cues can be penalized by the dialogue-density rule unless explicitly written as an ACTION beat.
- **Mid-Sentence Scene Boundaries:** Windows spanning scene cuts occasionally slice midway through a multi-beat sentence, requiring video audio-cut crossfades.

7. PRODUCTION SCALING & ECONOMICS

- Cost Efficiency: Brute-force LLM calls cost ~$1.50/ep ($150/season). Two-Stage Hybrid costs <$0.002/ep ($0.18/season) - a 99.8% cost reduction.
- High Throughput: Batch processes an entire 50-episode catalog in under 2 minutes (214 promos generated).
- Multimodal Expansion Roadmap: Feed rendered MP4s into Gemini 2.0 Flash Video API to score facial expressions and music swell alongside text.